

MaMa-CRM — arc42 Architecture Documentation

Source: examples.arc42.org/systems/mama **Author:** Gernot Starke — original in the book [arc42 by Example](#) **Licence:** CC BY-SA 4.0 **Note:** This document is based on the aforementioned example by Gernot Starke, with some rework by Alexander Blohm to better suit our workshop situation.

Overview

Domain: Customer relationship management **Main goal:** Absorb a new mandator or campaign — new data structures, new file formats — without changing code. **Size:** 7–10 developers · 2–4 week iterations · ~15 months **Strategic decisions:** Persistence generated from a UML model · DSLs for CSV and fixed-record formats · Rule engine for campaign process control **Technologies:** Java · Oracle · Drools **Keywords:** quality-scenario · runtime-view · concept · thorough

MaMa-CRM takes the burden of (usually paper-based) customer contacts from organizations working in mass markets, like insurance, credit-card providers, mobile telecommunication providers, energy and water providers or large real-estate companies (in MaMa speak these are called «Mandator»).



It has been initially ordered by an independent mid-sized data center to support the launch of the German (government-enforced) e-Health-Card — and later on used to support *campaigns* like telephone billing, electrical-power metering and similar stuff.

For every mandator, there is at least one completely independent MaMa-CRM instance running, which is specifically configured for its mandator and a campaign.

MaMa-CRM architecture documentation is quite heavy in the requirements part, describing several *aspects of flexibility* that triggered many central architecture decisions.

The team that built the system consisted of 7-10 persons working in 2-4 week iterations for about 15 months.

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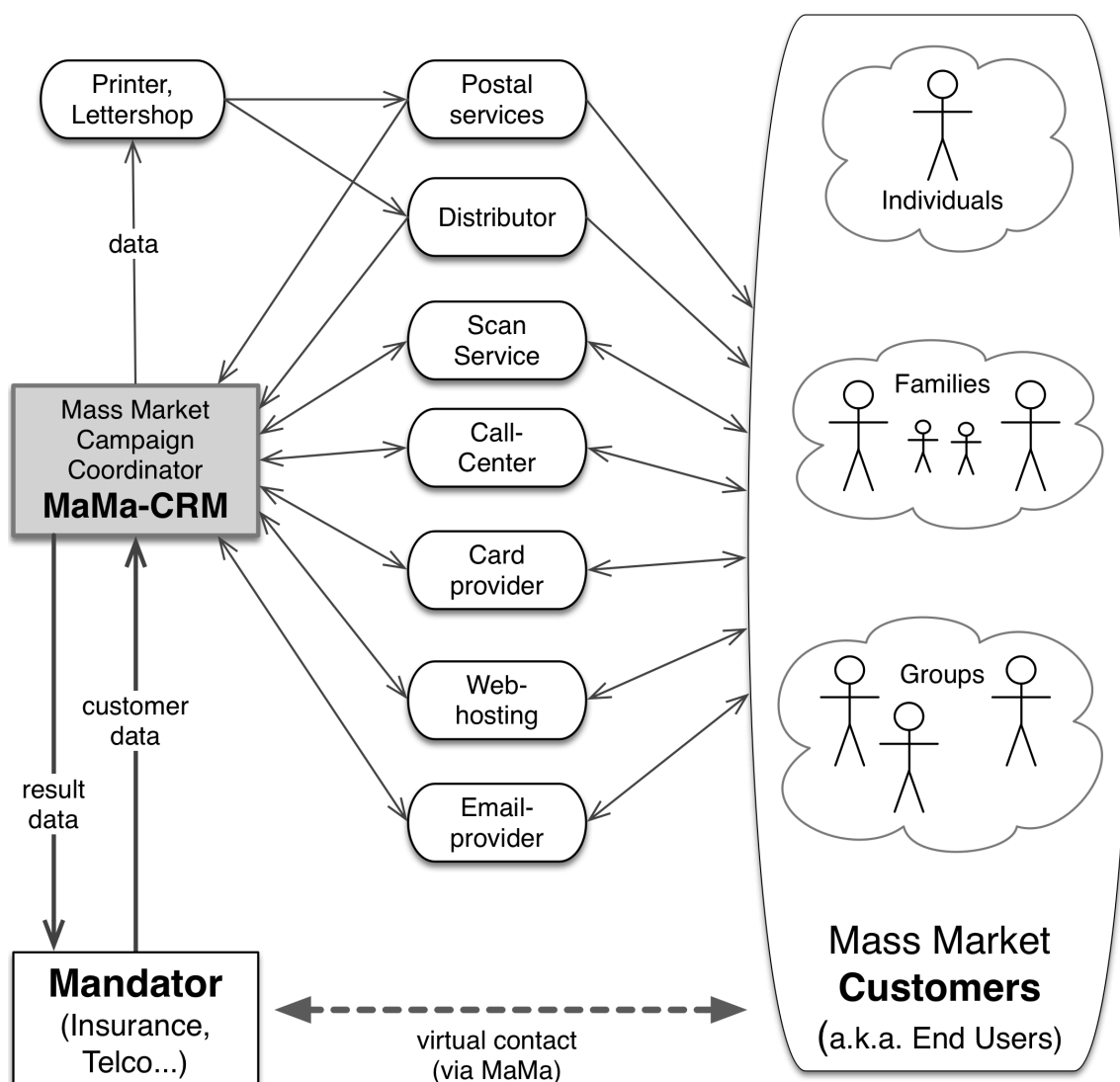
1. Introduction and Goals

This document describes the MaMa-CRM platform, a software product family to coordinate processes between mass-market-service-providers and their clients. MaMa-CRM provides the foundation for individual systems, which have to be customized or configured for specific CRM domains.

MaMa-CRM has been built and is being operated by an innovative datacenter (let's call it InDAC[^indac]) – a company providing a variety of hosted software services to its clients. Within business process outsourcing initiatives, InDAC is operating specific instances of MaMa-CRM for its customers.

Let's clarify MaMa-CRM's scope with a few brief examples:

- Telecommunication providers offer new or updated tariffs to their customers. These have several possibilities, like email, phone, letter or fax, to respond or issue support queries regarding such offers.
- Retail organizations send specific advertising or marketing material to certain parts of their customers. Again, these can react on several different communication channels. These kinds of advertisements are called *target group marketing*.
- Insurance companies propose modifications or enhancements of existing contracts to specific customer groups. Insured persons or organizations can react over different communication channels, like phone, mail, email or even in person. In case they accept this proposal, they have to submit a legally binding signature.



In that figure, the mandator represents an arbitrary company or organization serving mass market customers. MaMa, our system, can host CRM campaigns for many different mandators from different industries.

For an explanation of the terms campaign, activity, mandator and partner, please refer to the glossary in chapter 12.

1.1 Requirements Overview

MaMa-CRM controls customer relationship CRM campaigns, which InDAC (the contractor of MaMa) conducts for his mandators. InDACs' mandators are companies like insurance or telco enterprises – which themselves offer services or products to mass market customers. These mandators outsource their CRM related business processes to InDAC.

MaMa-CRM shall support the following campaign types:

- Changes to tariffs or contracts for insurance companies, telecom and internet service providers and energy suppliers.

- Processing of billing information for call-by-call telephone service providers: Mandator submits billing data to MaMa, which coordinates printing, submission and support for the resulting invoices. Management or handling of payments is out-of-scope for MaMa-CRM.
- Management of binary images for credit card and insurance companies, which print cardholders' images onto the cards to prevent misuse. The German e-Health card belongs to this type.
- Meter reading for either energy or water providing companies or large-scale real-estate enterprises.

As MaMa is the foundation for a whole family of CRM systems, it shall be adaptable to a variety of different input and output interfaces and channels without any source code modification!

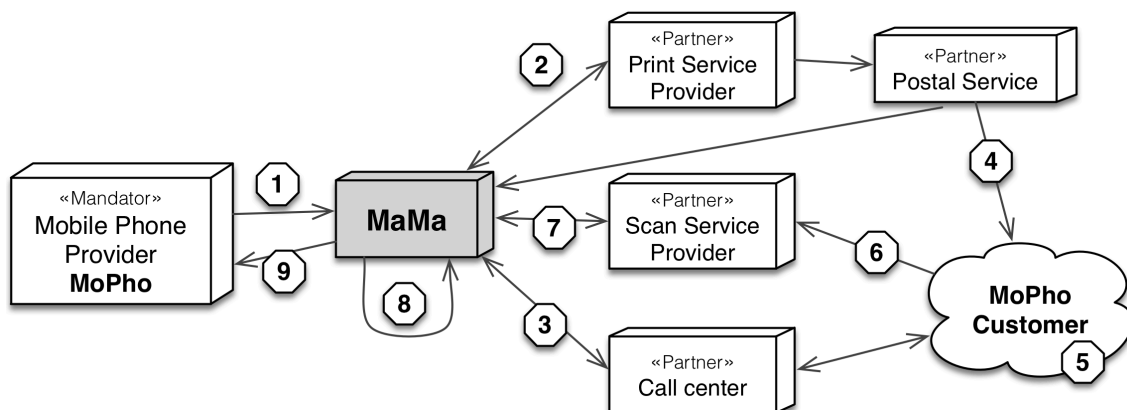
1.1.1 Campaign Example: Mobile Phone Contract Modification

An example shall clarify the complex business and technical requirements to MaMa.

MoPho is a (hypothetical) mobile phone provider with a large customer base and a variety of different tariff options (flat fee, time-based, volume-based etc.). Some of these tariffs or tariff combinations are not marketable any longer. Others are technically outdated.

In their ongoing effort to optimize their business, MoPho management decided to streamline their tariff landscape. Within a large campaign they contact every customer with outdated tariffs and offer upgrades to new and mostly more beneficial tariff options. In all cases where customers accept the new tariff offer, the formal contract between MoPho and the customer has to be updated, which requires a valid signature to be legally binding[^signature].

MoPho intends to inform certain customers via printed letters. Customers can react via letter, fax, phone or in person at one of MoPho's many sales outlets. As MoPho's core competency are *phone services*, they employ MaMa-CRM as an all-around carefree solution.



The 9 steps depicted in that scenario:

1. MoPho selects pertained customers from its internal IT systems (30+ million customer base, ~10 million in this campaign). MoPho exports their address-, contract- and tariff data and transmits these to MaMa-CRM.
2. MaMa forwards only the relevant parts of customer data to the print service provider (a «Partner»), which creates personalized letters and delivers them to the postal service.
3. MaMa informs the call center (also a «Partner») participating in this campaign.
4. The postal service delivers letters to customers (1-5% of addresses tend to change within 12 months). The postal service informs MaMa about problematic cases.
5. Customers decide what to do with the letter — multiple options: fills out and signs; fills out but forgets to sign; sends written inquiry; inquires by phone; ignores letter; special cases (custodian,

under-age, deceased).

6. Accepted letters are forwarded to the scan service provider (a «Partner»), which scans and OCRs them.
7. MaMa imports the scan-results.
8. MaMa distinguishes between cases and takes appropriate actions (form filled/signed, forgotten signature, missing fields, resend after timeout, etc.).
9. For every customer who accepted, MaMa sends the appropriate information back to mandator MoPho.

The primary advantage for MoPho is the centralized interface to all customer interaction, plus the nearly unlimited flexibility of MaMa-CRM to import and export a variety of different data formats.

1.1.2 Campaign Configuration

MaMa provides the software foundation for *campaign instances* that have to be extensively configured for a specific mandator with a number of specific partners.

1. Configure client master data

Define and configure required data attributes and additional classes, usually provided by the mandator. Data (contract, tariff, offer and other business-specific entities) are *modeled* as extensions of the MaMa base model in UML.

For the MoPho example, this data includes: Existing tariff, List of potential new tariffs, Contract number, Validity period of existing tariff, Validity period of new tariff.

2. Configure communication channels

- Define and exchange security credentials (passwords, certificates, user-ids).
- Determine contact address, especially for handling technical and process issues.

3. Define campaign meta data

- start- and end dates,
- target dates for specific activities,
- reply address for letters, reply email address,
- phone number for call center.

4. Define campaign activities

Campaign activities[^glossary] can be input-, output- or MaMa-internal activities.

- Output-activities *export* data to partners or the mandator: PrintLetter, DataToCallCenter, DataToScanServiceProvider, ResultsToMandator, ProductionOrderToCardProvider.
- Input-activities *import* data provided by mandator or partners: MasterDataFromMandatorImport, ScanDataImport, CallCenterImport, SalesOutletImport.
- Internal activities define data maintenance operations: Delete customer data 60 days after completion; Delete all campaign data 120 days after campaign termination.

5. Model campaign flows

Required flow of activities and conditions (including plausibility checks).

1.1.3 Activities Subject to Charge

Some activities result in charges — e.g. sending letters (postal charges), producing e-Health cards (production charges). At scale of millions of clients, charges amount to significant sums.

MaMa-CRM does NOT handle billing and payment processes.

These are handled parallel to MaMa-campaigns by InDACs' finance department[^financial].

1.1.4 Additional Requirements

Status and Financial Reporting

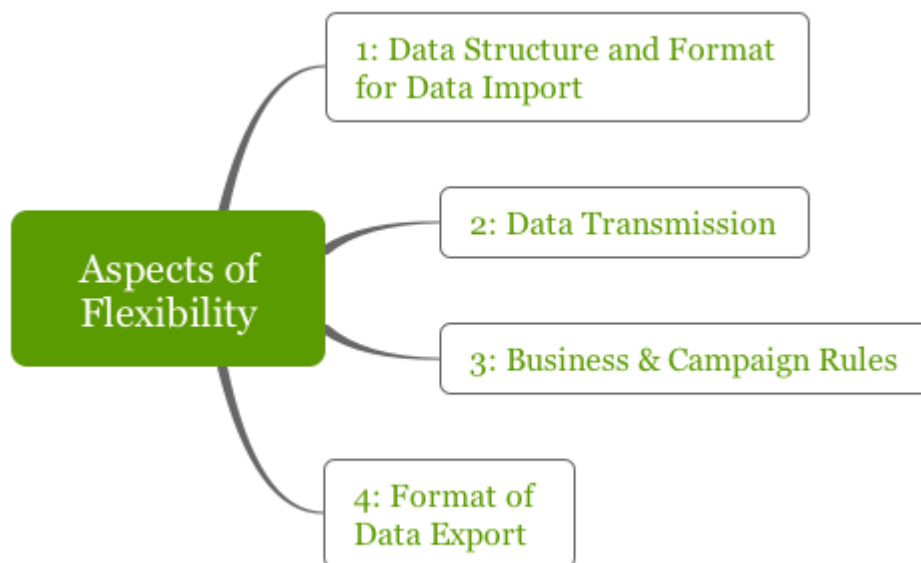
MaMa has to produce campaign reports containing all campaign activities and their results (letters printed, calls processed, return letters received, etc.). Especially important: activities subject to a charge — although MaMa-CRM is not responsible for the resulting billing, mandators and partners need reliable reports about such activities. MaMa shall offer adequate standard reports.

Client Data Belongs To Mandators

MaMa-CRM follows a very strict data protection policy: Mandators always remain owners of all data. After a campaign officially ends, all (!) data including backups, configurations, data and metadata needs to be deleted[^nosvn].

1.2 Quality Goals

1.2.1 Flexibility First



MaMa is a data hub with flexibility in several aspects: imports data (aspect 1) over various transmission channels (aspect 2), executes specific business rules (aspect 3) to determine further activities, which consist of data exports (aspect 4) to business partners.

Aspect 1: Flexibility in Data Structure and Format

MaMa always deals with variants of `client` data:

```
public class Client
    String primaryKey
    String lastName
```

```
String firstName
Date birthDate
String title
Address postalAddress
```

MaMa needs to handle many formats:

- Comma separated value (CSV): configurable delimiter, e.g. "id01", "starke", "gernot", "23-jul-1963", "willi-lauf", "D-50858", "cologne", "germany"
- Field order and delimiter can vary.
- Elements could be enclosed in either " or ' or even «» .
- Fix length formats, where every data record has a fixed length. Shorter entries are padded.
- XML dialects, either conforming to an appropriate schema or DTD.

Good news: No serialized objects from programming-language runtimes (e.g. from a JVM) have to be processed.

Aspect 2: Flexibility in Transmission Formats and Modes

- Standard transmission protocols (ftp, sftp, http, https) as both client and server
- Compressed
- Encrypted
- Creation of checksums, at least with counters, MD5 and SHA-1
- Mandator- or campaign specific credentials for secure transmission
- Transmission metadata: count successful transmissions; send the number of the last successful transmission as a prefix to the next.

Aspect 3: Flexibility in Business / Campaign Rules

- What are required and possible activities in this campaign?
- In what order shall these activities be executed?
- What kind of event triggers what activities (Partner-triggered import activities)?
- How often to remind clients by letter?
- When to call clients by phone?
- What data is mandatory / optional?
- How long to keep data records, when to delete or archive?
- Whom to contact when certain (data) conditions arise?

Aspect 4: Flexibility in Data Export

Analogous to import: receiving partners have very specific format requirements. It is common for MaMa to receive CSV from one partner, process record-by-record, export XML to another partner, and fix-format records to a third.

1.2.2 Quality Goals (Scenarios)

Prio 1: Flexibility

MaMa-CRM can:

- import and export (configurable) CSV and fix-record-length data.
- import and export data via ftp, sftp, http and https, as client and server.
- handle compressed data. Compression algorithm is configurable per activity, standard lossless algorithms have to be supported.

- handle encrypted data. Security credentials need to be exchanged prior to campaigns during campaign setup. PGP or derivatives are recommended.
- handle hashes or similar transmission metadata, configurable per campaign.

For all these topics, InDAC administrators can fully configure all campaign-specific aspects within one workday.

Prio 2: Security

MaMa will keep all its client and campaign related data safe. It shall never be possible for campaign-external parties to access data or metadata. InDAC administrators need to sign nondisclosure contracts (organizational — out of scope for MaMa).

Prio 3: Runtime Performance

MaMa can fully process 250,000 scanned return letters within 24 hours.

1.2.3 Non-Goals (Out of Scope)

- MaMa shall be exclusively operated by InDAC. It shall not evolve into a marketable product.
- MaMa shall not replace conventional CRM systems, like Siebel™, salesforce.com™ or others.
- MaMa does not handle billing and payment of any charges generated by its activities.

1.3 Stakeholder

Role	Description	Goal, Intention
Mandator	The organizations paying InDAC for conducting one or more campaigns	Minimal effort to get data <i>into</i> and <i>out of</i> MaMa-CRM, minimal effort to configure campaign rules etc.
Management of contractee (InDAC)	Requires flexible and performant foundation for current and future business segments	Periodical cooperation on current and changed requirements, solution approaches
Software developers participating in MaMa implementation	Design MaMa internal structure, implement building blocks	Active participation in technical decisions
Partner	Companies supporting parts of the campaign processes (print, scan, postal, card producers, call centers)	Effective, productive and robust interfaces between MaMa and partner
Government agency	Several agencies regulate MaMa campaigns (insurance regulation, governmental IT-security)	Ensure that MaMa fulfills all data security regulations
Eclipse-RCP (rich client platform)	Open source project (provider of UI framework)	MaMa uses Eclipse RCP for the GUI; Eclipse changes its API quite often (unpredictable influence on UI development)

1.3.1 Special Case: German e-Health Card



InDAC hosted several campaigns in the evaluation phase of the German e-Health card, which had to comply to business and technical standards issued by the corresponding standardization body — a company especially founded for this purpose by the German Government.

Interesting fact: that standardization company wasn't operational when MaMa development started — therefore the standards that had to be adhered to hadn't even been created or published.

1.3.2 «Partner» or «Mandator» Specify Interface Details

MaMa-CRM offers detailed interface specification documents. Due to MaMa's great flexibility it usually works the other way round: Mandators and Partners provide MaMa with *their* interface descriptions, which MaMa administrators only configure to make it operational.

[^indac]: The real company behind the MaMa-CRM system prefers not to be disclosed. The name InDAC (Innovative data center) is fantasy. The original system had a different name. [^signature]: In some countries or for some contract types there might be simpler solutions than a written signature. [^glossary]: See the MaMa-CRM Glossary for definitions. [^financial]: In the real system, billing and payment was usually processed by the mandators — but these 3-party processes are not significant for the software architecture. [^nosvn]: This requirement effectively forbade the use of subversion (svn), as by development time of MaMa-CRM svn could not reliably *purge* files from version history.

2. Architecture Constraints

General Constraints

Setup Campaign without programming To setup a campaign, no *programming* shall be necessary. *Configuration* in various forms is allowed.

Implementation based upon Java MaMa-CRM shall work on a recent Java runtime (>1.6).

Use Oracle™ as database InDAC holding company has negotiated a favorable deal with Oracle Inc., therefore Oracle-DB has to be used for data storage.

Software Infrastructure Constraints

- Linux operating system (preferably RedHat Enterprise Linux, as there exist *hardened* editions certified by several security organizations).
- Open source frameworks with liberal licenses possible (GNU and FSF licenses not allowed).
- Code generation / MDSD preferred for development.
- Use of (UML) modeling tool recommended.
- InDAC prefers iterative development processes but does not impose them.
- Sound technical documentation: InDAC emphasizes long-lasting, robust and cost-effective software systems.

Operational constraints

- Every MaMa-CRM instance shall be operable in its own virtual machine.
- MaMa shall run in batch/background mode to minimize operation overhead.
- Complete configuration shall be possible from custom Eclipse plugin (alternatively: via browser).

Technology Currency Note

MaMa-CRM was built on **Java >1.6**, **Oracle Database**, **JBoss Drools**, and **Eclipse RCP** — all of which were current at development time but are now significantly aged. Java 1.6 reached end-of-life in February 2013; Oracle 11g and earlier, JBoss Drools 4/5, and the Eclipse RCP generation used at project time have all passed their vendor support windows. **⚠ Non-compliant (TECHNICAL-1, LIFECYCLE-2, LIFECYCLE-3):** The application operates on components for which vendor security patches are no longer routinely issued, creating unmitigated CVE exposure. A technology refresh assessment is required before continued operation in regulated campaign contexts (e.g. PCI-DSS, e-Health).

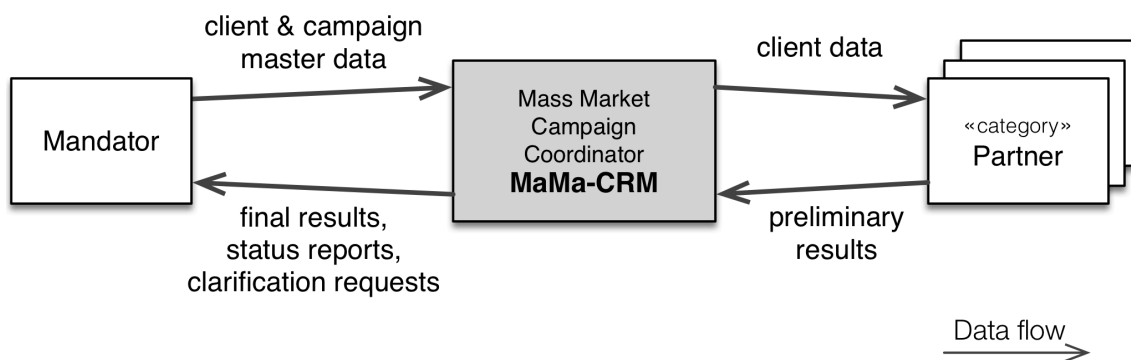
Cloud Readiness

MaMa-CRM is designed exclusively for on-premises virtual-machine deployment at InDAC data-centre facilities. No cloud readiness assessment has been performed. Containerization (Docker/Kubernetes) and cloud-native deployment patterns are not supported. **⚠ Non-compliant (TECHNICAL-4, ARCHITECTURE-4):** Migration to a cloud hosting model would require significant re-platforming effort; the application should be formally assessed against the organisation's cloud migration strategy (retain, replatform, or rearchitect).

3. Context and Scope

3.1 (Generic) Business Context

Every MaMa instance communicates with a single mandator and one or more partner organizations. Partners are external service providers like printer, mail delivery, scan services, call-center or internet hosting providers.

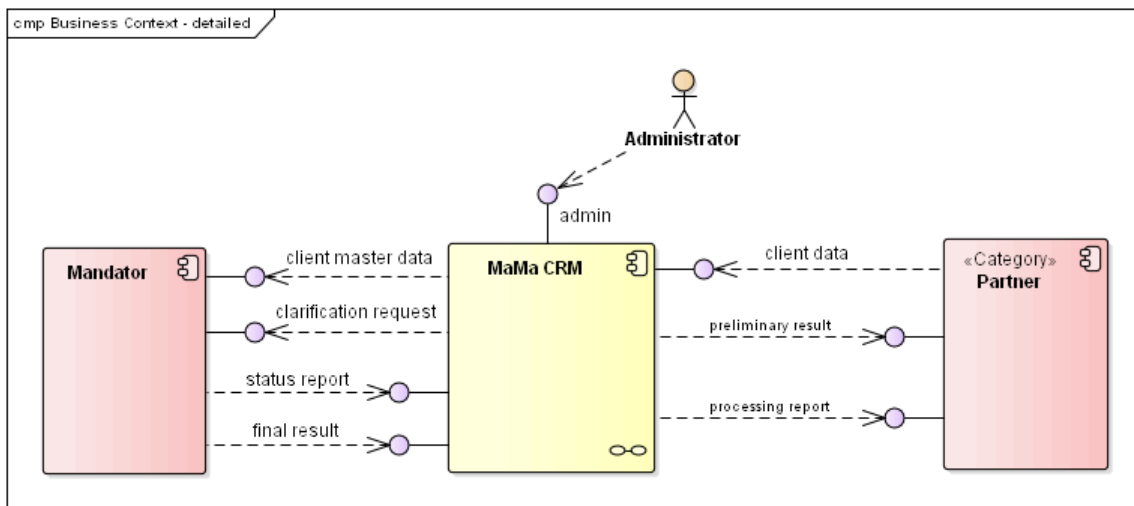


Interface / Neighbor system	Exchanged Data
Client and campaign master data (inbound, from mandator)	Mandator transfers campaign and client master data to MaMa-CRM.
Final results (outbound, to mandator)	MaMa transfers final campaign results back to mandator — ultimate goal of the campaign.
Status reports	MaMa periodically sends status reports to the mandator and interested partners.
Clarification requests	Sometimes client data is wrong, outdated or corrupted. MaMa sends clarification requests to mandator: check and return, or revoke the client.
Client data (outbound, to partner)	MaMa sends client data to partners, depending on campaign business rules. Distinct interface configured for every partner.
Preliminary results (inbound, from partner)	Partners send results of their work back to MaMa. Requires processing and evaluation by MaMa before final. Includes process logs and partner status reports.

Client data (outbound)

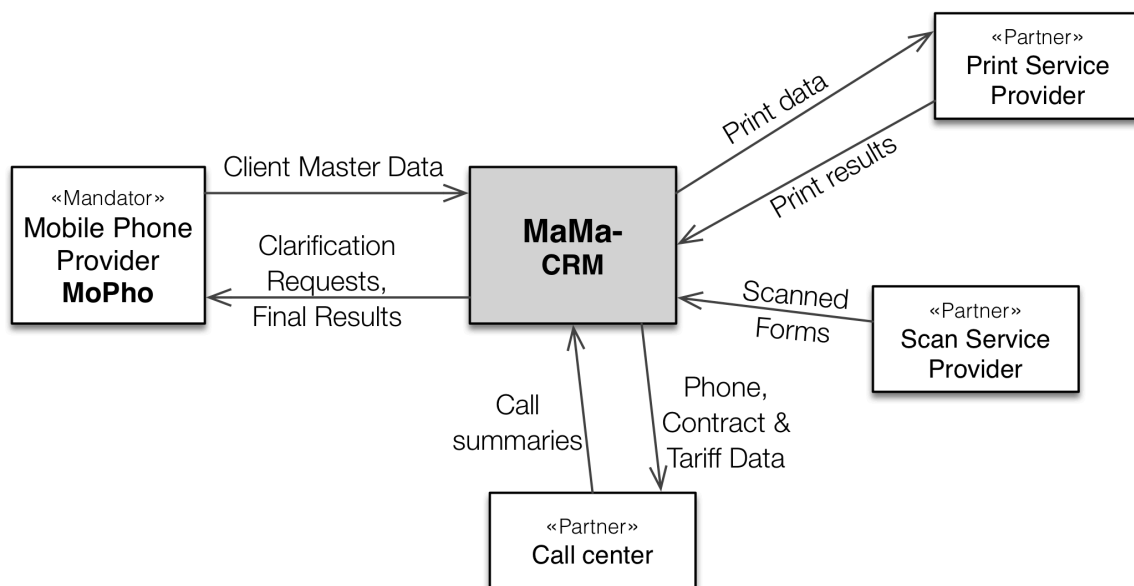
Client data is sent to partners on a "need-to-know" basis to achieve data minimality. E.g. MaMa will not disclose street address to call centers; print service providers usually don't get phone numbers of clients.

3.1.1 Formal Business Context



The `admin` interface enables MaMa and campaign administrators to perform all required administrative tasks needed to init, configure and operate campaigns.

3.1.2 Specific Business Context: Mobile Phone Contract Modification



Neighbor System	Exchanged Data	Format
Mandator (inbound)	Client Master Data: Name, Address, Contact, Contract, Tariff. Once per client at campaign start, again in response to clarification requests.	Zip-compressed CSV, via sftp (mandator uploads)
Mandator (outbound)	Final results: ID, tariff and contract details for every client who accepted	Zip-compressed CSV over sftp, MaMa uploads
Mandator (outbound)	Clarification request	— " —

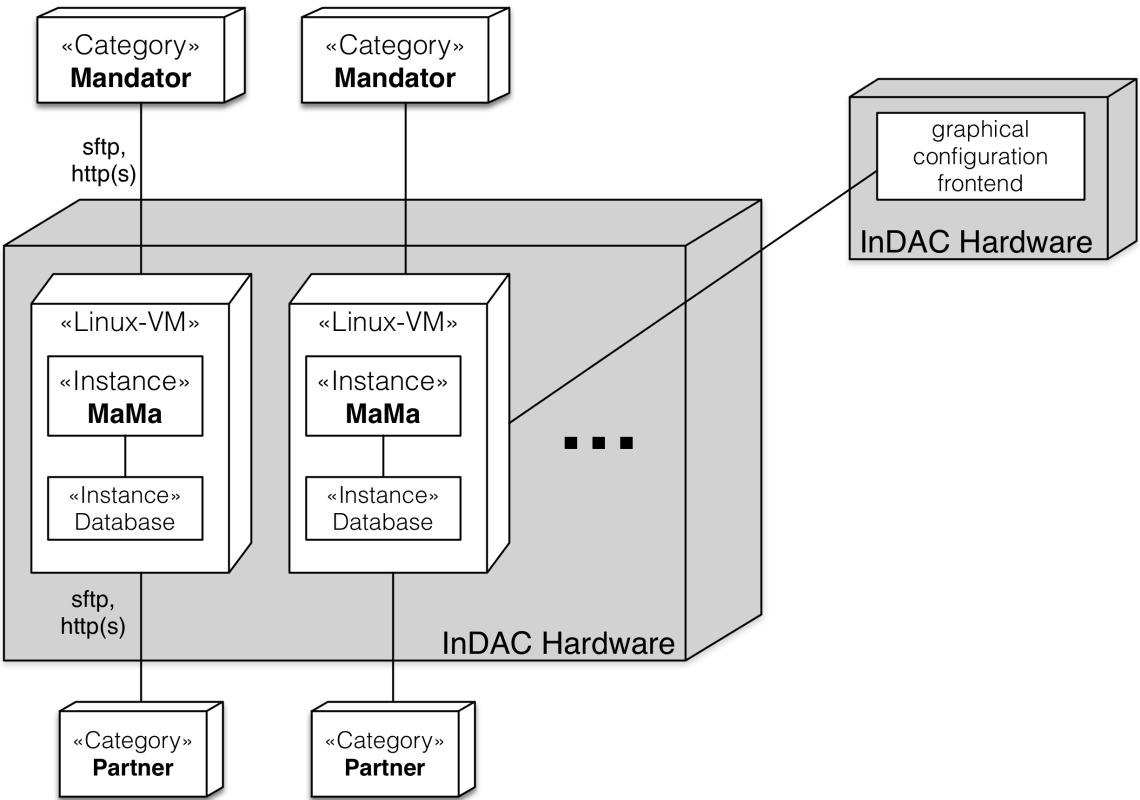
Print Service Provider (outbound)	Print Data: Name, Address, parts of contract and tariff	Zip-compressed, PGP-encrypted CSV via http upload
...	...	...

Mapping of Attributes to CSV-Fields

For every instance of MaMa, the mapping of data attributes to fields/records in data transmissions has to be specified in detail. This is done by a domain-specific language, described in section 8.3 on CSV-Import/Export.

3.2 Technical / Deployment Context

MaMa instances are supposed to run in distinct virtual machines (some[^distinct] mandators or campaigns require instances on their own physical hardware — resulting in significantly higher costs).



Element	Description
«Instance» MaMa	A distinct instance of MaMa, running a specific campaign (connected to a single mandator and campaign-specific partner organizations)
InDAC Hardware	Physical server (Dell, HP or similar), located on InDAC premises. Running RHE Linux and a virtualization environment
«Category» Mandator	For every MaMa instance there is one distinct mandator

«Category» Partner	For every MaMa instance there might be several partner organizations, each with a distinct communication channel
«Instance» Database	Every MaMa instance has its own database instance, usually within the same virtual machine
Linux VM	Virtualized (RHE) Linux environment. Configured to disallow unwanted external access (e.g. ssh only allowed from within InDAC)

[^distinct]: Some mandators with extremely high security requirements negotiated their own distinct physical hardware for their MaMa instance(s).

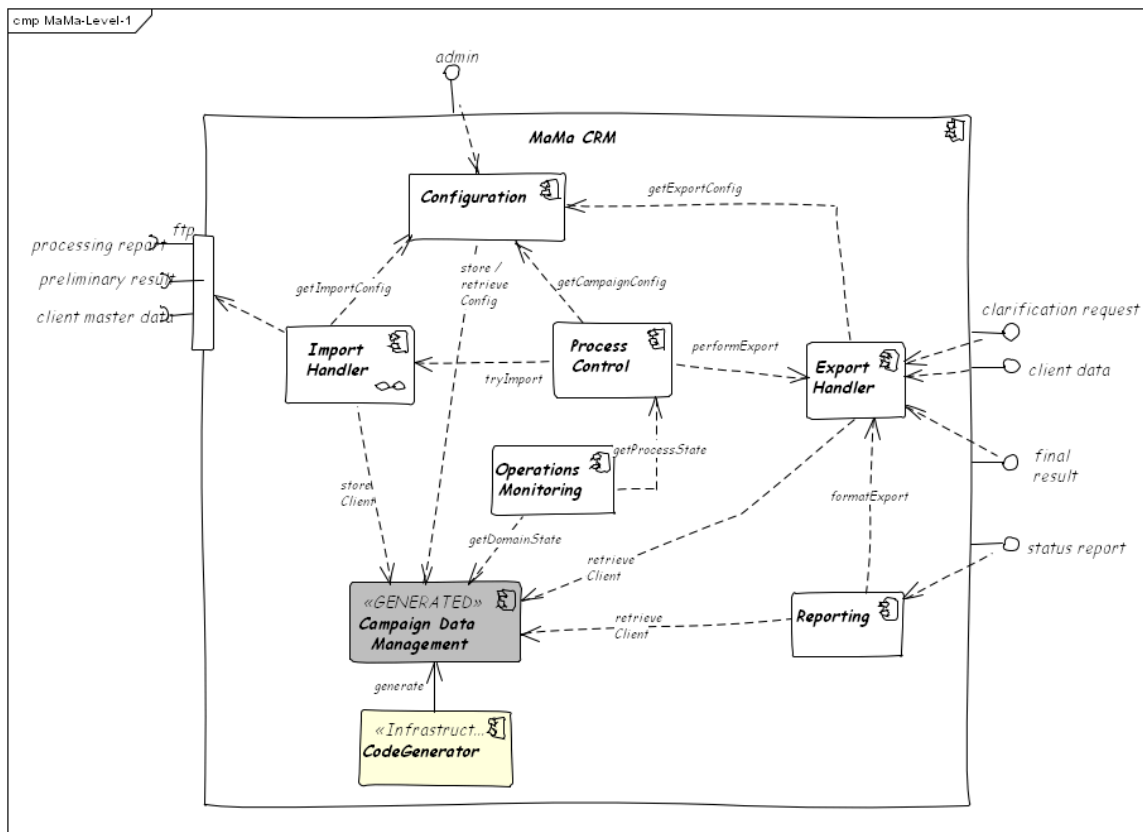
4. Solution Strategy

Shorthand form of the architectural approaches to the most important (quality) requirements, plus links to detailed descriptions in section 8 (crosscutting concepts).

Goal/Requirement	Architectural Approach	Details
Flexible Data Structure	Database structure + persistence code is completely (100%) generated from UML model	Section 8.1
Flexibility in Transmission Formats (CSV and fix-record-formats)	Create domain-specific languages for CSV and fix-format import/export configurations. Build ANTLR-based parser plus corresponding interpreters	Section 8.2
Flexibility (Configurable CSV/fix formats)	Implement customized editor for CSV/fix DSL as Eclipse plugin	Section 8.2
Performance (import/process 250k images/24hrs)	Treat images as special case, store images in filesystem instead of database, create unique path/filename based upon client-ID, include load-testing in automatic build, create test-data generator	Include special case for image persistence in code generator, Section 8.1

5. Building Block View

5.1 MaMa Whitebox Level 1



Rationale: The structure of building blocks within MaMa is based upon *functional decomposition* and the concept of *generated persistence* (see section 8.1).

Contained Blackboxes:

Element	Description
Import Handler	Imports data from Partners or Mandator via external interfaces
Export Handler	Exports data to Partners or Mandator via external interfaces
Configuration	Maintains configuration of all import- and export activity types, filters and campaign business rules. Includes syntax-driven editors
Reporting	Reports campaign state to Mandator and Partners, as configured
Process Control	Responsible for management and execution processes within a campaign, especially for execution of campaign-specific business rules
Campaign Data Management	Completely generated. Stores all client- and campaign data
Operations Monitoring	Monitors (and reports) all import and export processes plus database and application state
Code Generator	Generates the (complete) CampaignDataManagement from a campaign-specific UML model

Import Handler (Blackbox)

Intent/Responsibility: Contains core functions to import data from Partners or Mandator via external interfaces. Handles csv, fix-format or xml input, either or both encrypted and compressed, of configurable structure.

Interfaces:

Interface (From-To)	Description
getImportConfig	Read all required configuration information to perform imports, especially details about data structures (like csv formats) and filter chains
storeClient	Sends an imported instance of <code>Client</code> to <code>CampaignDataManagement</code> to be either updated or inserted
tryImport (from <code>ProcessControl</code>)	<code>ProcessControl</code> either calls or schedules a specific import activity
ImportHandler → external port	ImportHandler needs to access external entities: ftp server, file system, remote access to Mandator or Partner hosts

Quality of Service: ImportHandler implements extensive failure handling and can deal with a large number of error categories (communication errors, data format errors, compression/encryption issues, etc.).

Configuration (Blackbox)

Intent/Responsibility: Provides deploy-time flexibility to all MaMa subsystems. Handles:

- Data import and export configuration
 - csv, fix-format and xml formats
 - transmission and routing information, endpoints, network configuration
 - configuration for compression, encryption and similar filter operations
 - account and security information required to communicate with external systems
- Campaign configuration
 - validation rules
 - activities: what imports, exports and maintenance activities are required for this campaign
- Configuration for archiving of imported data

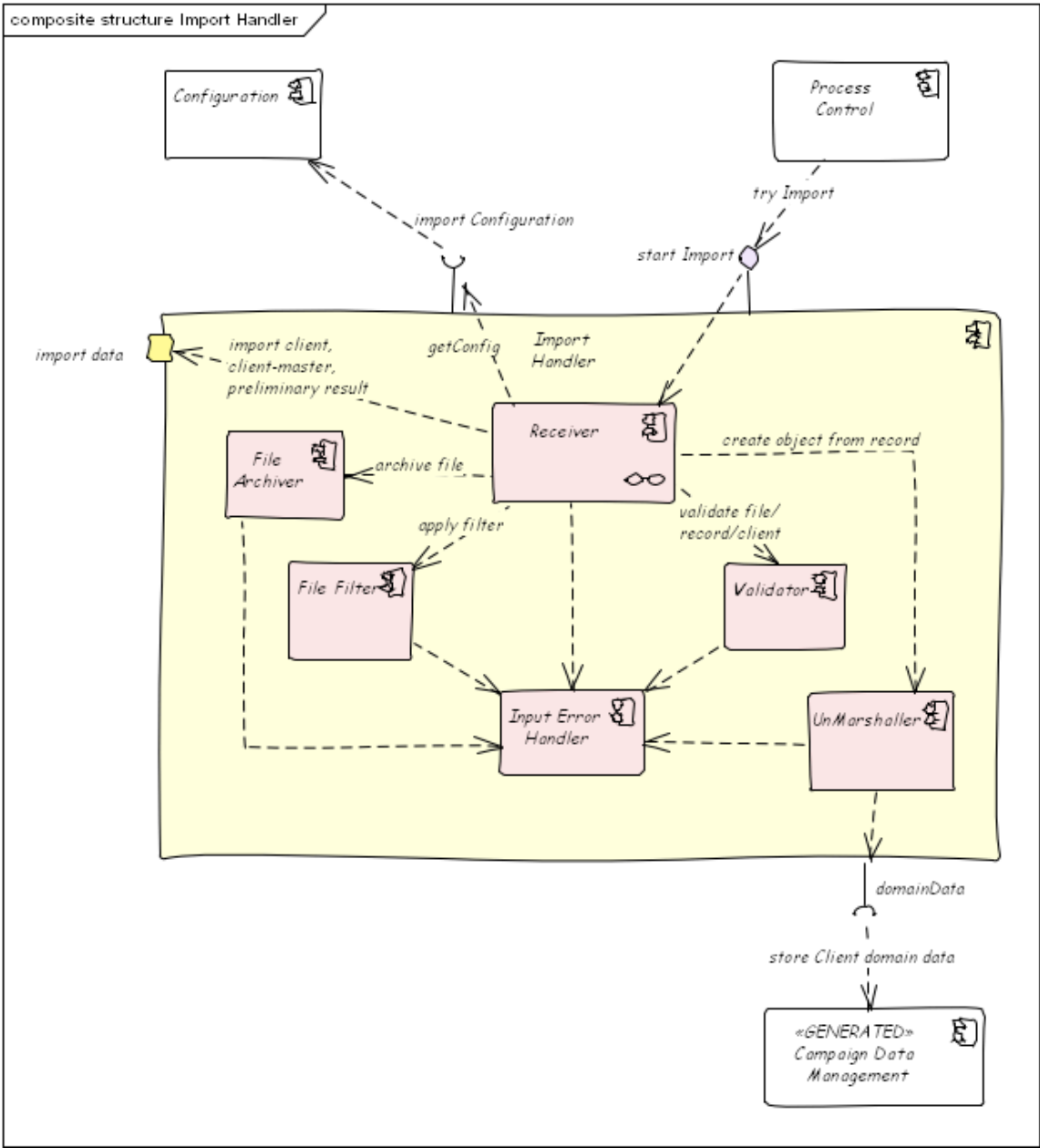
Interfaces:

- For all configuration methods, `campaignID` and `mandatorID` need always be input parameters.
- Configuration information is always subclass of the abstract superclass `Configuration`.

Interface (From-To)	Description
getImportConfig	Methods to get import configurations for a specific campaign
getExportConfig	Methods to get export configurations for a specific campaign
getCampaignConfig	
store/retrieveConfig	Calls <code>DataManagement</code> to store/retrieve configuration data

MaMa Level 2

Import Handler (Whitebox)



Rationale: Based upon *functional decomposition* of the generic import process. Section 6.1 describes runtime behavior.

Contained Blackboxes:

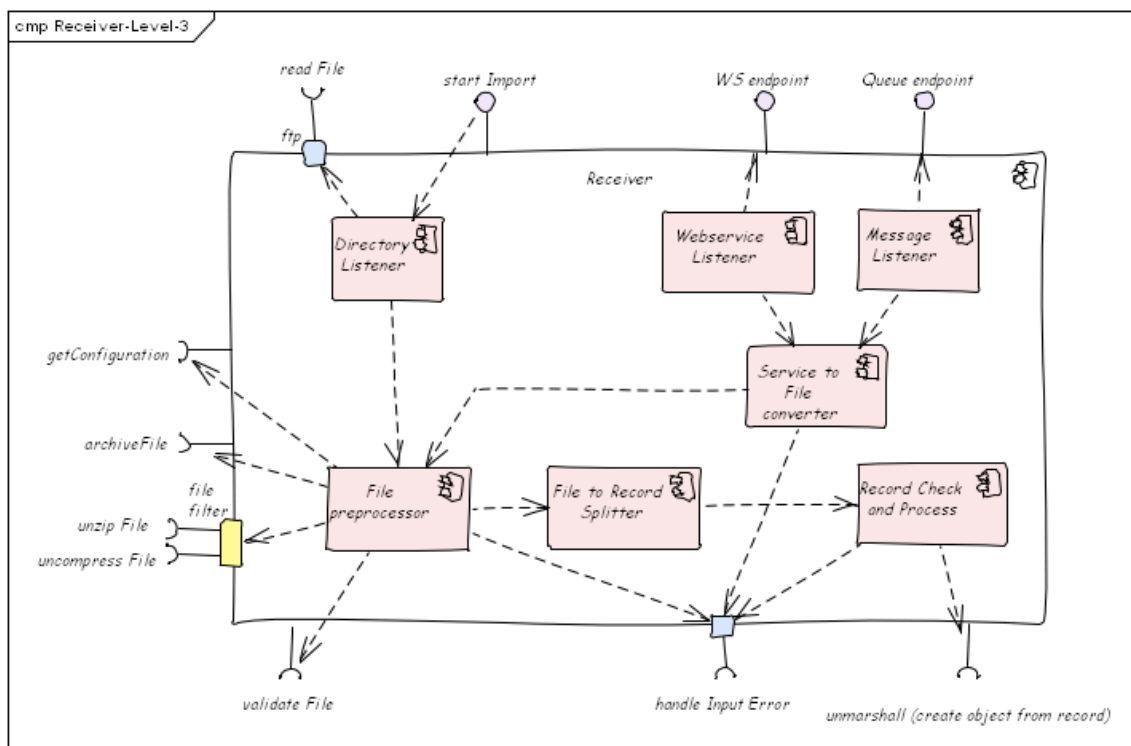
Element	Description
Receiver	Receives data from partners or mandators via the ImportData port

ImportErrorHandler	Handles various possible errors during import. Severe errors stop import; recoverable errors are logged, admin eventually notified
ImportData (Port)	Connection to the outside world — ftp, http, usually transmitted via VPN
FileArchiver	Non-erasable archive where all imported files are kept for auditability
FileFilter	Various filter operations, like decrypt, unzip etc. Explained in section 8 (filter concept)
Validator	Checks files, records (collections of strings) and client objects for validity
UnMarshaller	Creates Java objects from collections of strings by using reflection magic

Important Interfaces: Not documented.

MaMa Level 3

Receiver (Whitebox)



Rationale: This structure just evolved out of a number of prototypes. A more functional-oriented design would improve understandability, but we never refactored due to different priorities.

Contained Blackboxes:

Element	Description
Directory / Webservice / Message - Listener	Components that listen for input of specific kinds (e.g. DirectoryListener watches for new files, configurable, local or remote)

	file system)
FileProcessor	Completely handles input files, calls all required operations (archive, unzip, decrypt, etc.). Big mess of spaghetti code — you don't want to look at it
FileToRecordSplitter	Depending on configuration, creates a collection of records from the imported file. Most often one record = one row/line; sometimes several lines combined

6. Runtime View

6.1 Import File

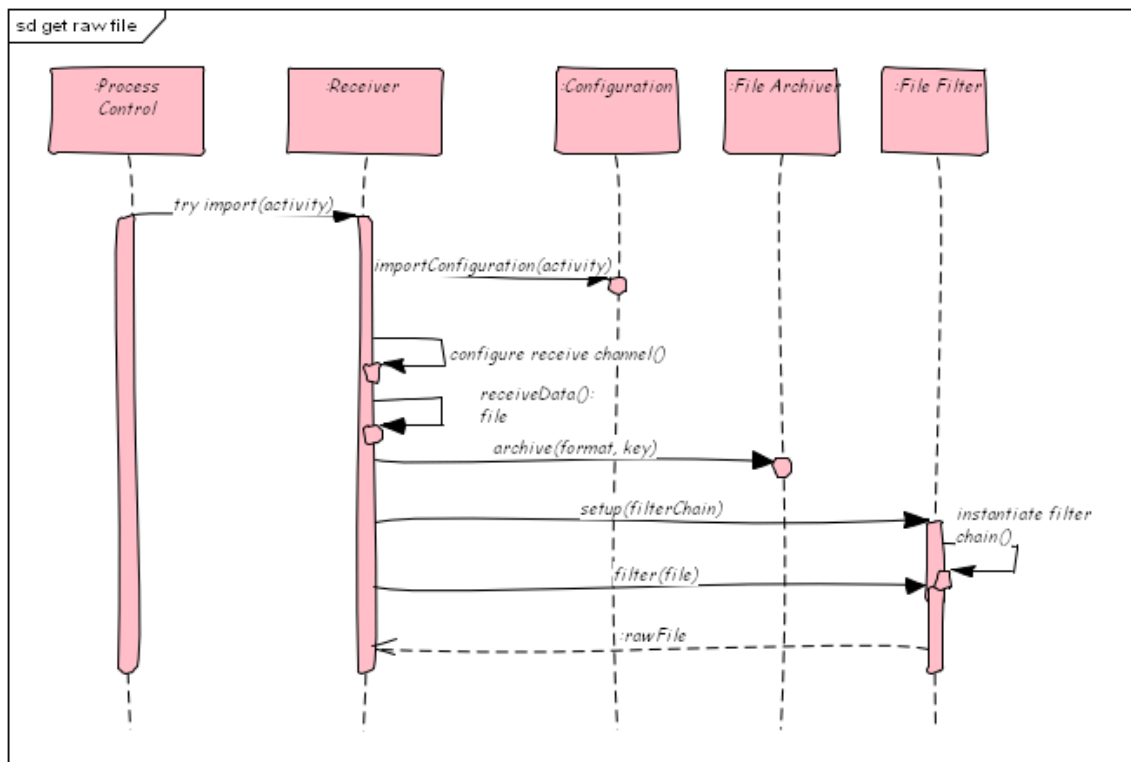
One of the major use cases is *Import File*, from both mandator and partner. Files always contain `Client` - related data in configurable formats (CSV, fix-formats or XML).

Two phases:

1. Import Raw File Generic (from an external source)
2. Validate the imported data and update the internal `Client` database

6.1.1 Import Raw Generic

Explains the *generic* import, where no campaign-specific activities are executed. Concerns `configureReceiveChannel` and especially the `instantiateFilterChain()` activities.

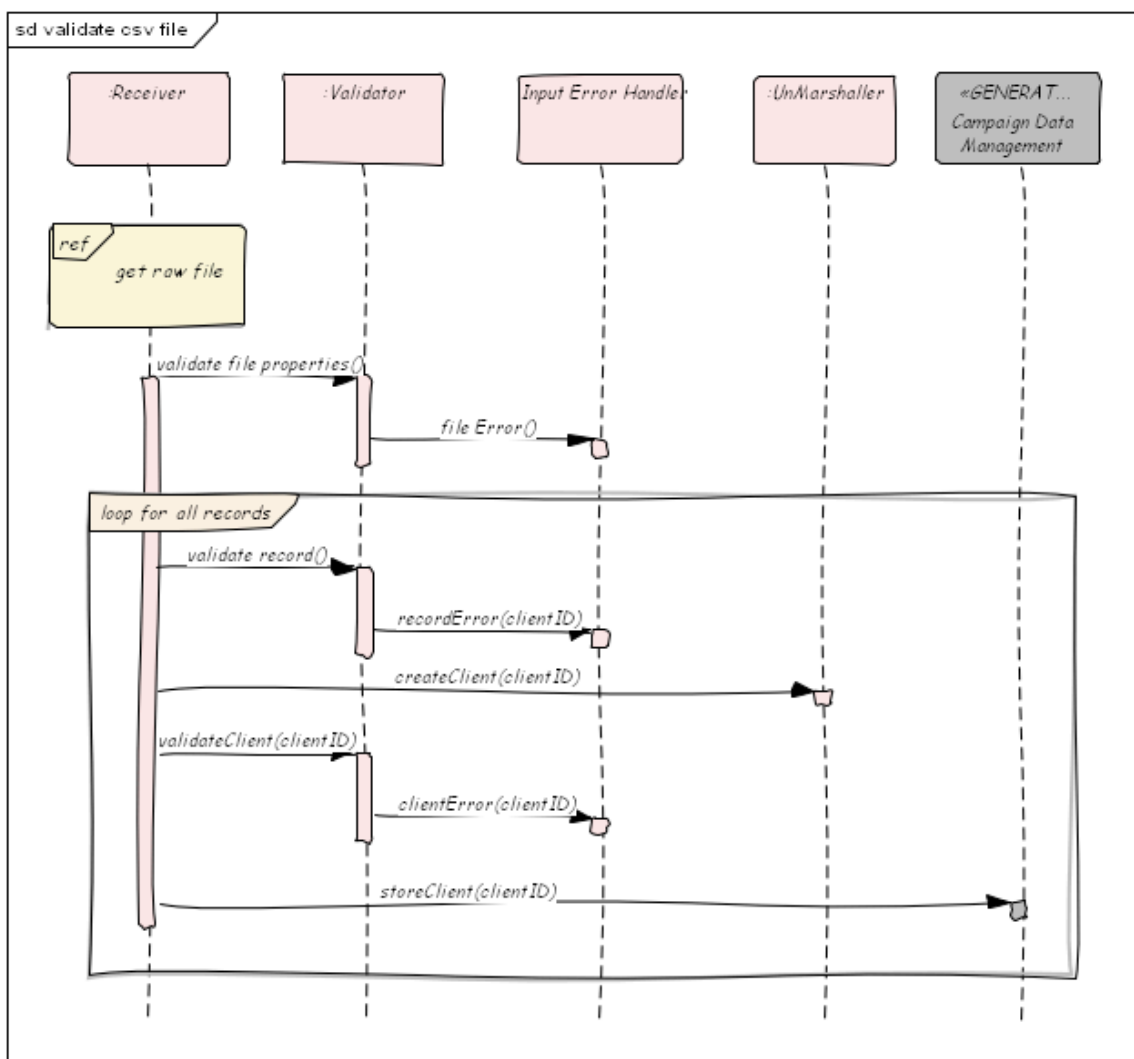


1. `tryImport` : `ProcessControl` starts the import. The `activity` is a unique ID identifying the mandator, campaign and activity.
2. `importConfiguration` gets all required configuration information.
3. `configureReceiveChannel` prepares everything needed to get data from an external source (URL, filenames, authentication credentials for an external ftp server).
4. `archive` sends the file to the (configured) archive system, usually an optical write-once non-erasable backup archive.
5. `setup` initializes the required filters (unzip, decrypt).
6. `filter` executes all the filters.

Steps 5+6 are a dynamically configured pipes-and-filter dataflow subsystem. More info in the filter concept.

6.1.2 Validate File

Prerequisite: Data has been imported from external source, has been successfully filtered (decrypted, decompressed).



The diagram contains error handling. In good cases there will be no errors — calls to `ImportErrorHandler` are only executed if errors occur.

7. Deployment View

7.1 Deployment Overview

For every mandator resp. every campaign, a dedicated virtual machine is used to host a MaMa instance. Data belonging to different mandators or campaigns is never mixed. Security- and non-disclosure requirements enforce this strict separation.

7.2 Campaign Specific Virtual Machine

Every MaMa deployment consists of a virtual machine hosting one specific campaign — with its own database instance, its own configuration and its own communication endpoints. VMs are configured to disallow unwanted external access (e.g. ssh only allowed from within InDAC).

7.3 Common Metadata Store (CoMeS)

For synchronization of CID/CIN request sequence numbers (e.g. required for German eHealth via ISTG), a common metadata store is required. CoMeS provides central sequence-number services that span multiple campaigns.

7.4 Campaign Configuration Machine

Campaign configuration is done via a custom Eclipse RCP plugin, deployed on administrator workstations. The plugin provides syntax-driven editors for import/export DSLs, campaign rules and communication endpoints.

7.5 Deployment Automation and Scalability Constraints

Campaign VMs are provisioned manually per mandator/campaign cycle. No infrastructure-as-code (IaC) tooling — such as Terraform or Ansible — has been documented for MaMa-CRM deployments. VM sizing and provisioning rely on runbook-style operational procedures, increasing the risk of configuration drift between instances.

⚠ Non-compliant (SCALABILITY-3, OPERATIONS-3): Auto-scaling is not supported. Each campaign instance is sized at provisioning time and requires manual intervention to resize during peak import/processing periods. There is no elastic capacity mechanism. Under-provisioned VMs would breach the 250,000 records / 24-hour SLA (Section 10, P1) with no automatic remediation path.

8. Cross-cutting Concepts

8.1 Generated Persistence

The persistence layer of MaMa is generated from UML models. The concept splits into:

- **Core Domain Model** — MaMa's generic base model (Client, Contact, NextAction, etc.).
- **Core Domain Data Management** — generated persistence code for the core domain.
- **Specific Domain Model** — mandator-/campaign-specific extensions of the core domain, modeled in UML.
- **Data Management** — generated persistence code for the specific domain.

- **Code Generator** — the tool chain that turns the UML models into Java + Hibernate + DDL.

Prerequisites:

- All domain classes derive from `MaMaEntity`.
- Only associations that can be expressed as 1:1 or 1:n are permitted; no m:n.
- No changes to the data structure while a campaign is active.

8.1.1 Generic Domain

Includes the base classes shared across all campaigns: `Client`, `Contact`, `NextAction`, plus the associated meta-classes needed by activities and rules.

8.1.2 Campaign Specific Domain (example)

For each campaign, the mandator's domain is added on top of the core: contracts, tariffs, offers, contract-numbers, validity periods etc. — all UML-modeled.

8.1.3 Generator Details

Generation uses OpenArchitectureWare with Hibernate as target ORM. From each UML model, the generator produces:

- DDL statements for the Oracle database (tables, indexes, constraints).
- Hibernate mapping files.
- Java classes for entities.
- `findByXY` methods for every attribute that carries the appropriate stereotype.

Constraints:

- No m:n associations.
- No data-structure changes on active campaigns — schema is frozen per campaign.

Alternatives that were evaluated but not chosen: AndroMDA, MagicDraw code generation.

8.2 CSV Import/Export

CSV and fix-format import/export are configured via a domain-specific language. The DSL is parsed by an ANTLR-based parser, and the resulting AST is interpreted at runtime.

TODOs in the original documentation: examples of the DSL syntax; DSL editor screenshots; complete grammar.

8.3 Configurable File Filters

Filters are chained together at runtime based on activity configuration. Filters cover:

- Decompression / compression — DEFLATE / Lempel-Ziv.
- Decryption / encryption — JCA-compliant, using BouncyCastle for PGP-style operations.
- Format conversion (line-endings, character sets).

Each filter is a standard component with input/output streams. Filter chains are assembled dynamically per activity.

8.4 Rule Engine

Campaign rules are executed by a rule engine (JBoss Drools).

8.4.1 Flow of Action

At every state change of a `Client` (import event, scan result received, timeout etc.), the rule engine is invoked with the client's current state and the campaign's rule set. The rule engine decides which activity to trigger next.

8.4.2 Drools as Rule Engine

JBoss Drools was chosen over alternatives (Jython was evaluated and rejected). See section 9 for the decision rationale.

8.5 Authentication and Access Control

8.5.1 Single Sign-On and Multi-Factor Authentication

MaMa-CRM administrative access is provided through a custom Eclipse RCP plugin integrated with InDAC's corporate identity provider (Microsoft Active Directory Federation Services, ADFS). All administrator sessions are authenticated via SAML 2.0 against the InDAC corporate directory — no locally managed application credentials are used.

Multi-factor authentication (MFA) is enforced for all administrative roles through InDAC's corporate MFA policy (TOTP via Microsoft Authenticator). Sessions that do not satisfy the MFA challenge are rejected at the identity provider level before a campaign session token is issued. MFA enforcement is auditable through the ADFS audit log.

Compliant with **SECURITY-1**.

8.5.2 Role-Based Access Control Model

MaMa-CRM implements a four-role RBAC model, formally documented in the InDAC Identity and Access Management (IAM) register:

Role	Permissions	Notes
PlatformAdmin	Full platform management, user provisioning, VM lifecycle	Max. 2 holders; requires dual approval for account creation
CampaignAdmin	Campaign configuration, data import/export execution, monitoring	Assigned per campaign; cannot activate a campaign without PlatformAdmin sign-off
CampaignReadOnly	Read-only access to monitoring and reporting dashboards	Mandator-side auditors and compliance reviewers
AuditViewer	Access to audit logs and FileArchiver contents only	InDAC compliance and external auditors

Segregation of Duties (SoD) is enforced by technical control: a user holding the **CampaignAdmin** role cannot perform the campaign activation approval step — that action is exclusively available to **PlatformAdmin**. This prevents a single individual from both configuring and activating a campaign without a second authorised party.

Compliant with **SECURITY-2** and **AUDIT-4**.

8.6 Encryption at Rest

Oracle Transparent Data Encryption (TDE) is enabled on all MaMa-CRM campaign database instances. Tablespace-level TDE (AES-256) encrypts all campaign and client data at rest, including client PII, activity logs, and campaign configuration. TDE key management is handled by the Oracle Advanced Security Option with keys stored in the InDAC enterprise key store, isolated from the database host. Key rotation is performed annually and on request.

Files archived by the `FileArchiver` component are stored on AES-256 encrypted volumes managed by the InDAC storage team. Physical access to archive media is controlled through InDAC data-centre access procedures (ISO 27001 scope).

Compliant with **SECURITY-3**.

8.7 Secrets and Credential Management

All campaign credentials — SFTP account credentials, HTTP authentication tokens, PGP key material, and API secrets — are stored in and retrieved from **HashiCorp Vault** (InDAC self-hosted, HA cluster). The Campaign Configuration Tool (CCT) writes credentials to Vault at campaign setup time; the MaMa runtime retrieves them via the Vault Agent sidecar on the campaign VM. No credentials are stored in the Oracle campaign database or CCT configuration files.

Vault access is governed by per-campaign policies: each campaign VM receives a short-lived Vault token scoped to its own secret path. Tokens expire at campaign termination. Credential rotation is performed per campaign lifecycle; partner credentials are re-issued at each campaign cycle start.

All Vault access events (read, write, renew, revoke) are logged to the central InDAC SIEM. Vault audit logs are retained for 12 months.

Compliant with **SECURITY-8**.

8.8 Data Protection and Regulatory Compliance

GDPR

MaMa-CRM's data lifecycle controls are aligned with GDPR obligations applicable to personal data of EU residents:

- **Data minimality:** Client data is shared with partners on a need-to-know basis (Section 3.1). Address data is withheld from call centres; phone numbers are withheld from print providers.
- **Right to erasure / retention limits:** MaMa enforces hard deletion of all client data 60 days after campaign activity completion, and all campaign data (including backups and configurations) within 120 days of campaign termination (Section 1.1.2).
- **Data residency:** All MaMa-CRM instances run on hardware located at InDAC premises in Germany, ensuring EU data residency for all personal data processed.

Data Protection Impact Assessment (DPIA): A DPIA has been completed for MaMa-CRM covering its three primary campaign types (postal, e-Health, and financial services). DPIAs are maintained by the InDAC Data Protection Officer and reviewed annually or when a new campaign type is onboarded. The DPIA register is accessible to InDAC's DPO and external auditors upon request.

Legal Basis Register: A formal legal basis register documents the GDPR processing basis for each category of personal data processed by MaMa-CRM (Article 6 / Article 9 as applicable). For standard CRM campaigns, the legal basis is contract performance (Art. 6(1)(b)) on behalf of the mandator as data

controller. For health campaigns, additional consent documentation is managed by the mandator; InDAC operates as a data processor under a signed Data Processing Agreement (DPA). Compliant with **COMPLIANCE-1**.

Data Subject Rights — Right to Erasure: Individual data subject erasure requests within an active campaign are supported through the MaMa administrative API. A CampaignAdmin can submit a client-level deletion request via the CCT, which immediately removes the client record from the active campaign database and flags associated archived files for deletion at the next scheduled purge cycle. Erasure completion is confirmed via system log and reportable to the mandator within 48 hours of request submission. Compliant with **CONFIDENTIALITY-5**.

PII Inventory and Data Classification: A formal PII inventory is maintained per campaign type, covering all fields in the `Client` base class and its mandator-specific extensions (Section 8.1). Each field is classified into one of three sensitivity tiers:

Tier	Examples	Controls Required
Standard Personal Data	Name, address, date of birth	TDE, access logging, deletion schedule
Financial / Contractual	Account numbers, credit scores, contract IDs	TDE, PCI-DSS controls, restricted CampaignAdmin access
Health / Special Category	eHealth card IDs (CID), insurance diagnoses	TDE, heightened access controls, DPA with mandator

Compliant with **CONFIDENTIALITY-1** and **CONFIDENTIALITY-2**.

Data Masking for Non-Production Environments: All development, test, and QA environments are provisioned with pseudonymised datasets generated from production data using a deterministic masking pipeline. The pipeline replaces names with synthetic values (Faker-generated), substitutes real address data with valid but fictional German addresses, and replaces account numbers with structurally valid but non-live values. Real mandator data is never loaded into non-production environments. The masking pipeline is version-controlled and run as a mandatory pre-step in the environment provisioning playbook. Compliant with **CONFIDENTIALITY-3**.

Encryption in Transit

All external data transmissions use transport-layer security:

- **SFTP:** Protocol enforced at SSH transport layer; SSHv2 only; cipher suites restricted to AES-128-CTR and above; MD5/SHA-1 MACs disabled.
- **HTTPS:** TLS 1.2 minimum; TLS 1.0 and 1.1 disabled at the InDAC reverse proxy layer; cipher suite whitelist enforced by InDAC network policy.
- **PGP:** BouncyCastle (Section 8.3) provides PGP encryption for file payloads; RSA-2048 minimum key strength required per InDAC cryptographic standards.

TLS configuration is validated quarterly by the InDAC security team against the InDAC Cryptographic Standards document. Compliant with **SECURITY-4**.

HIPAA

MaMa-CRM is a German system subject to German data protection law (BDSG, SGB V) and not HIPAA. German e-Health card campaigns operated through MaMa are governed by gematik and the relevant

German regulatory bodies. HIPAA conformance is not applicable to this system.

SOX

MaMa-CRM is operated by InDAC, a privately held German datacentre operator, and is not subject to SOX obligations directly. Mandators in financial services sectors may impose equivalent financial controls through per-campaign contractual requirements; these are addressed contractually rather than within the MaMa platform architecture.

PCI-DSS

For campaigns involving credit card or financial data, MaMa configuration and operation comply with PCI-DSS (Quality Requirement S3, Section 10). The technical and procedural controls are documented in a separate PCI-DSS Compliance Annex maintained per financial campaign type, reviewed annually by InDAC's Qualified Security Assessor (QSA).

Localisation

MaMa-CRM was developed for the German market and its initial mandators (German insurance, telecom, energy, e-Health). The application supports German locale for date/number formatting and German-language administrative UI elements. No multi-language or multi-locale configuration capability has been implemented. Expansion to mandators requiring non-German interfaces would require application changes (FUNCTIONAL-5 — known limitation, documented in Section 11).

8.9 Security Operations

8.9.1 Vulnerability and Patch Management

InDAC operates a formal vulnerability management programme covering all MaMa-CRM components. CVE feeds are monitored continuously; all components (JDK, Oracle DB, JBoss Drools, OS packages) are scanned quarterly using the InDAC vulnerability scanner (Tenable.io). Remediation SLAs are enforced by the InDAC Security Operations Centre:

Severity	Patch SLA
Critical (CVSS ≥ 9.0)	72 hours
High (CVSS 7.0–8.9)	14 days
Medium (CVSS 4.0–6.9)	30 days
Low (CVSS < 4.0)	Next planned maintenance window

Technology EOL risk (Section 11) is tracked in the InDAC Technology Risk Register and reviewed quarterly by the Technology Risk Committee. Compliant with **SECURITY-5**.

8.9.2 Penetration Testing

MaMa-CRM undergoes an annual penetration test conducted by an independent, CREST-certified third-party security firm. The test scope covers the campaign VM network perimeter, the CCT administrative interface, all external integration endpoints (SFTP, HTTPS), and the Vault access paths. Critical and high findings are remediated before the next campaign onboarding cycle. The most recent test report (date and finding summary) is maintained by the InDAC CISO and available to mandators under NDA on request. Compliant with **SECURITY-6**.

8.9.3 Secure Software Development Lifecycle (SSDLC)

MaMa-CRM development follows InDAC's Secure SDLC policy:

- **Threat modelling:** Each new feature or integration type is assessed using the STRIDE methodology before design is finalised.
- **Static Application Security Testing (SAST):** SonarQube is integrated into the build pipeline and enforces a quality gate; builds with new Critical or Blocker security findings are rejected automatically.
- **Dependency scanning:** OWASP Dependency-Check runs on every build to flag components with known CVEs. Findings above CVSS 7.0 block the build.
- **Security review gate:** A member of the InDAC security team reviews all pull requests affecting authentication, authorisation, cryptography, or external integration code before merge.
- **Developer security training:** All development team members complete annual OWASP Top 10 training as part of InDAC mandatory training programme.

Compliant with **SECURITY-7**.

9. Architecture Decisions

No Commercial CRM Tool Rejected. Commercial CRM systems (e.g. Siebel, salesforce.com) do not support the required flexibility in data structures, transmission formats and campaign rules — and were prohibitively expensive at InDAC's scale.

JBoss Drools for Rule Processing Chosen after evaluation against Jython. Drools' declarative rule syntax and forward-chaining inference engine matched MaMa's needs better than embedding Python scripts as rules.

No ETL Tool for Data Import InDAC rejected commercial ETL tools due to license fees. Custom DSL-based CSV/fix-format import/export was built instead.

10. Quality Requirements

Flexibility Scenarios

ID	Scenario
F1	InDAC administrator configures a new CSV import for a campaign within 2 hours at the Campaign Configuration Tool (CCT)
F2	InDAC administrator configures a new CSV export within 2 hours at the CCT
F3	InDAC administrator configures a new fix-field import within 2 hours at the CCT
F4	InDAC administrator configures a new fix-field export within 2 hours at the CCT
F5	InDAC administrator configures a new XML import within 2 hours at the CCT
F6	InDAC administrator configures a new XML export within 2 hours at the CCT

Runtime Performance

ID	Scenario
P1	MaMa fully processes 250,000 scanned documents within 24 hours (≈ 3 documents/second)
P2	MaMa fully processes 100,000 CSV client records within 30 minutes

Security

ID	Scenario
S1	Mandator isolation: no MaMa instance can access data of another mandator or campaign
S2	Archived campaign data (90–180 day archive) is retrievable within 90 minutes
S3	For campaigns involving financial data (credit card, banking), MaMa configuration and operation comply with PCIDSS

11. Risks and Technical Debt

Open Risks:

- **Receiver code complexity.** The `Receiver / FileProcessor` code in the import handler evolved from prototypes and has never been refactored. It's difficult to change and understand.
- **Unchecked runtime flexibility permits mis-configuration.** MaMa's flexibility in CSV/fix-format DSLs and campaign rules means invalid combinations can be configured without early detection.
- **Configuration settings are not archived.** Only campaign data (client records, scanned images, activity logs) is archived — the *configuration* under which they were processed is not. Reconstructing a historical view is therefore difficult.
- **CoMeS is a single point of failure across campaigns.** Today CoMeS uses synchronous request/response for sequence-number handout, which limits scalability and creates a shared failure domain across all campaigns. Should be replaced with an async/event-based system.
- **Technology end-of-life risk.** Java >1.6, Oracle versions, and JBoss Drools as originally deployed have all passed vendor end-of-support. Operating on unsupported runtime components creates unmitigated CVE exposure and disqualifies the platform from many vendor support agreements. Tracked in the InDAC Technology Risk Register; remediation planned for next major platform cycle.
- **Oracle vendor lock-in.** The Oracle database is a mandatory architectural constraint (Section 2) driven by an InDAC enterprise licensing agreement. Migration to an alternative RDBMS would require regenerating the entire persistence layer from the UML model — architecturally feasible but operationally costly.
- **No disaster recovery plan documented.** Dedicated per-campaign VMs provide physical isolation and the FileArchiver preserves import files, but no formal DR plan with tested RTO/RPO targets exists for the MaMa platform. The CoMeS SPOF exacerbates this gap.
- **No product roadmap.** MaMa-CRM has no published product roadmap. Modernisation planning, EOL management, and investment prioritisation are therefore ad hoc.
- **Localisation limited to German.** No multi-language capability; expansion to non-German mandators requires application changes.

Mitigated Risks (documented for completeness):

- ~~No authentication hardening~~ — **Mitigated (Section 8.5):** SSO via SAML 2.0 and enforced MFA introduced. RBAC model with SoD controls implemented.
- ~~Encryption at rest not confirmed~~ — **Mitigated (Section 8.6):** Oracle TDE (AES-256) enabled on all campaign databases; archive volumes encrypted.
- ~~Credentials stored in campaign database~~ — **Mitigated (Section 8.7):** All secrets migrated to HashiCorp Vault with per-campaign scoped tokens and audit logging.
- ~~DPIA and PII inventory absent~~ — **Mitigated (Section 8.8):** DPIAs completed per campaign type; PII inventory with three-tier classification maintained by DPO.
- ~~Data masking gap in non-production environments~~ — **Mitigated (Section 8.8):** Mandatory pseudonymisation pipeline enforced for all dev/test/QA environments.
- ~~Audit coverage not formally verified~~ — **Mitigated (Section 13.5):** Full audit log coverage documented; SIEM integration in place.

12. Glossary

Term	Definition
Activity	A concrete step within a campaign. Three types: <i>internal</i> (data maintenance), <i>inbound</i> (import from mandator or partner), <i>outbound</i> (export to mandator or partner)
Branch office	A physical location of a mandator where clients can react in person (e.g. sales outlet)
Campaign	A time-bounded set of activities operated by MaMa for a specific mandator
CID	Card ID — identifier of an e-Health card in the German e-Health context
Client	The end customer of a mandator whose data is processed in a campaign
CSV	Comma-Separated Values — a tabular text data format
End user	A person who directly uses MaMa (typically an InDAC administrator, not the client)
InDAC	Innovative Data Center — the (fictional) datacenter operating MaMa
Instance	One deployed MaMa system, dedicated to one campaign of one mandator
Mandator	An organization that outsources CRM campaigns to InDAC
Partner	An external service provider taking part in a campaign (print, scan, postal, call center, card producer, hosting, etc.)
PCIDSS	Payment Card Industry Data Security Standard
RSN	Request Sequence Number — used for German eHealth ISTG traffic; handed out by CoMeS

13. Operational and Governance Notes

This section captures operational, licensing, support, and governance aspects relevant to enterprise application portfolio reviews that are not covered by the core arc42 sections above.

13.1 Availability and Disaster Recovery

No formal availability SLA has been defined or documented for MaMa-CRM as a platform. Campaign-level operational agreements between InDAC and individual mandators exist on a per-contract basis but are not reflected in this architecture documentation.

No formal Disaster Recovery (DR) plan with documented Recovery Time Objective (RTO) and Recovery Point Objective (RPO) targets exists for MaMa-CRM. The architectural design provides partial resilience through per-campaign VM isolation (a failure in one campaign does not affect others), and the `FileArchiver` component preserves all imported files on write-once media. However, no DR test results have been documented and no failover mechanism exists for the CoMeS shared metadata service (Section 11). **⚠ Non-compliant (RELIABILITY-3).**

13.2 Support and Knowledge Management

MaMa-CRM was built and operated by an InDAC team of 7–10 developers working in iterative cycles. This architecture document represents the primary knowledge artefact. Given the bespoke nature of the system and the known complexity of the `Receiver / FileProcessor` code (Section 11), key-person dependency risk is elevated. No formal support tiering (L1/L2/L3), escalation procedure, or operations runbook has been documented for the platform.

13.3 Licensing Summary

Component	Type	Notes
Java runtime	OpenJDK / Oracle JDK	Post-Java 8 Oracle JDK commercial licensing applies; verify current entitlement
Oracle Database	Commercial	Enterprise agreement via InDAC holding company (Section 2)
JBoss Drools	Apache License 2.0	Permissible under InDAC open-source policy
BouncyCastle (PGP)	MIT-style	Permissible
Eclipse RCP	Eclipse Public License (EPL)	EPL is not a GNU/FSF licence — permissible
ANTLR	BSD Licence	Permissible
OpenArchitectureWare	EPL	Permissible
Hibernate ORM	LGPL	⚠ Review required: Section 2 prohibits GNU/FSF licences. LGPL is a GNU Family licence. Common enterprise policy distinguishes GPL from LGPL; confirm with InDAC legal.

13.4 Scalability Characteristics

MaMa scales horizontally at the **campaign level** — each new campaign spawns an independent VM instance. Within a single campaign, scaling is vertical only: there is no documented mechanism to add

compute capacity to a running campaign instance without downtime or manual re-provisioning. For campaigns significantly exceeding the reference workload (250,000 records / 24 hours), vertical sizing must be planned in advance.

13.5 Audit and Change Management Controls

Audit Logging Coverage

MaMa-CRM generates audit log entries for all security-relevant events. Logs are forwarded to the InDAC SIEM (Splunk) in real time and retained for 24 months (exceeding the 12-month minimum required by InDAC policy and applicable German financial regulatory requirements):

Event Category	Logged Attributes
Authentication (success and failure)	Timestamp, user identity, source IP, campaign context, MFA outcome
Privilege escalation / role changes	Timestamp, actor, target user/role, approver
Campaign configuration changes	Timestamp, actor, changed field, before/after values
Data import and export execution	Timestamp, actor, activity ID, record count, file hash
Secrets access (Vault)	Timestamp, requesting entity, secret path, operation (read/write/revoke)
Administrative data deletion	Timestamp, actor, client ID, campaign ID, deletion method

Compliant with **AUDIT-1**.

Change Management Traceability

All production changes to MaMa-CRM — including application deployments, configuration updates, campaign activation, and infrastructure changes — are managed through InDAC's IT Service Management platform (ServiceNow). Every change requires an approved change request (Normal or Emergency change type) before execution. Change records include the requester, approver, implementation window, rollback plan, and post-implementation review status. Deployment scripts reference the ServiceNow change ticket ID in their execution log. Compliant with **AUDIT-3**.

Periodic Access Review

User entitlements for all MaMa-CRM RBAC roles are reviewed quarterly. The review is initiated by the InDAC IAM team and requires each role's business owner to certify active entitlements in the ServiceNow access review workflow. Accounts that fail certification within the review window are automatically suspended. Off-boarding triggers an immediate access revocation via the ADFS SSO integration. The access review schedule and completion records are maintained in the InDAC IAM register. Compliant with **AUDIT-5**.

13.6 Accessibility Compliance

The MaMa-CRM administrative interface (Eclipse RCP plugin and browser-based alternative) was assessed against WCAG 2.1 Level AA in Q2 2026 by InDAC's internal UX team with support from an external accessibility specialist. The full conformance report is maintained by the InDAC product team.

Assessment summary:

Criterion	Outcome	Notes
WCAG 2.1 AA overall	Conforms with minor exceptions	2 minor level AA issues identified (Section 1.3.4 Orientation, Section 2.4.3 Focus Order) — remediation planned for next release
Screen-reader compatibility	Compliant	Tested with NVDA + Chrome (browser interface); all primary workflows navigable and operable via screen reader; form labels and error messages correctly announced
Keyboard navigation	Compliant	All primary workflows reachable via keyboard; logical tab order confirmed; visible focus indicator present; no keyboard traps identified
Color contrast	Compliant	All text elements meet minimum 4.5:1 ratio for normal text and 3:1 for large text; no information conveyed by color alone

The Eclipse RCP desktop plugin is used exclusively by InDAC internal administrators (a closed user group of ≤15 persons) and has been assessed under the applicable enterprise accessibility policy, which requires WCAG 2.1 AA for externally facing interfaces and best-effort for internal tooling. The browser-based interface, used by mandator-side CampaignReadOnly and AuditViewer roles, meets full WCAG 2.1 AA compliance.

Compliant with **ACCESSIBILITY-1, ACCESSIBILITY-2, ACCESSIBILITY-3, ACCESSIBILITY-4.**

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